

ABDELRAHMAN ALAA ELDEEN EZZ ELDEEN

Generative AI Engineer | AI Agents | LLM Applications

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PROFESSIONAL SUMMARY

Generative AI Engineer with a strong Software Engineering background and experience building AI-powered applications, LLM-based systems, AI agents, intelligent validation pipelines, and enterprise software. Skilled in designing end-to-end AI solutions, integrating Large Language Models with APIs and business systems, and developing scalable applications using Python, Java, Spring Boot, FastAPI, React, and SQL.

TECHNICAL SKILLS

Generative AI & Agents: LLMs, AI Agents, RAG, MCP, Tool Calling, Prompt Engineering, LLM Integration, Prompt Injection Detection, AI Security & Validation, Ollama, Local LLMs, LLM Evaluation
Software Engineering: Python, Java, Spring Boot, FastAPI, REST APIs, API Integration, JWT, React, TypeScript, JavaScript
Data & Enterprise: MySQL, MariaDB, SQLite, Redis, Elasticsearch, Data Modeling, Data Governance, Metadata Management, Informatica Axon
Tools: Git, GitHub, Docker, Linux, Maven

EXPERIENCE

Generative AI Engineer

IDC Solutions | Full-time | Alexandria, Egypt | Jun 2026 - Present

- Design and develop AI-powered solutions and intelligent applications using Large Language Models and modern Generative AI technologies.
- Build AI agents and LLM-based systems integrated with backend services, APIs, and business workflows.
- Develop scalable AI pipelines with a focus on intelligent automation, validation, and enterprise integration.
- Combine Generative AI capabilities with strong software engineering practices to deliver reliable production-ready solutions.

Software Engineer

Jadara Solutions | Full-time | Alexandria, Egypt | Jul 2025 - May 2026

- Designed and developed an enterprise-grade Data Governance and Metadata Management platform based on industry-standard governance concepts and workflows.
- Gained hands-on experience with Informatica Data Governance tools, including Axon, and applied relevant governance and metadata management concepts in software development.
- Improved search and retrieval capabilities for large datasets through Elasticsearch configuration, tuning, and management.
- Developed backend services and REST APIs using Java and Spring Boot.
- Integrated Generative AI capabilities into enterprise applications to support intelligent automation and business workflows.

Java Programmer

Gomla Market Company | Full-time | Alexandria, Egypt | Jun 2020 - Oct 2020

SELECTED PROJECTS

AI Prompt Detection & Pre-Validation Layer

Designed a multi-layer security and validation pipeline for LLM and AI Agent systems, combining deterministic detection, prompt injection protection, policy enforcement, language and intent validation, LLM-based analysis, context management, and MCP/API integration.

AI Content Discovery & Research Agent

Developed an AI-driven pipeline for discovering emerging technology problems, analyzing solution gaps, progressively filtering research results, validating claims, and identifying high-value content opportunities.

Tailor ERP

Built a full-stack ERP system using Spring Boot, React, TypeScript, MySQL, and JWT. Developed inventory management, suppliers, stock movements, sales, invoices, payments, and reporting workflows.

AI Diabetes Detection System

Built a CNN-based image classification system and achieved 2nd Place in the Kufr El-Sheikh Olympics Competition.

EDUCATION

B.Sc. in Computer Science - Alexandria University | 2019 - 2023
Grade: Very Good (B+)

LANGUAGES

Arabic (Native) | English (Fluent)